Test Procedure for

DETERMINING THE COEFFICIENT OF THERMAL EXPANSION OF CONCRETE

TxDOT Designation: Tex-428-A

Effective Date: June 2023

1. SCOPE

1.1 Use this method to determine the linear coefficient of thermal expansion (CTE) of concrete by measuring the change in length of concrete cores or cylinders subjected to a range of temperatures.

1.2 The values given in parentheses (if provided) are not standard and may not be exact mathematical conversions. Use each system of units separately. Combining values from the two systems may result in non-conformance with the standard.

2. APPARATUS

2.1 *Water bath,* capable of maintaining temperatures of 50 ± 1°C (122 ± 2°F) and 10 ± 1°C (50 ± 2°F) with the water level maintained at least 1 in. (25.4 mm) above the support frame and fully submerged LVDT.

2.2 *Temperature measuring device (RTD or thermistor),* accurate to 0.1°C (0.2°F) and with a minimum resolution of 0.01°C (0.02°F), in appropriate location to measure water bath temperature.

2.3 *Caliper, comparator, or other suitable device,* used to measure the specimen length to the nearest 0.005 in. (0.1 mm).

2.4 *Support frame,* as shown in Figure 1.

2.4.1 The frame should be made of non-corroding material or make use of a sacrificial anode to minimize the effects of submersion in a water bath.

2.4.2 The frame should be rigid and fixed at an appropriate height to accommodate concrete cylinders trimmed to 7.0 ± 0.1 in. (178 ± 2.5 mm).

2.4.3 The frame should include a threaded mount and lock nut for adjusting the LVDT.

2.4.4 The base plate should have at least three support buttons equally spaced to evenly support a nominal 4 in. (100 mm) diameter concrete specimen.

2.5 *Concrete saw,* capable of sawing the ends of a cylindrical specimen perpendicular to the axis and parallel to each other.

2.6 *Fully submersible LVDT,* should have a minimum resolution of 0.00001 in. (0.00025 mm) and a range of ± 0.2 in. (± 5 mm). The end of the LVDT should be rounded or spherical.

2.7 *Reference specimen,* made of 304 stainless steel or other non-corrosive material of known thermal expansion, 7.0 in. ± 0.1 in. (178 mm ± 2.5 mm) in length and at least a 2 in. diameter.

2.8 *Verification specimen,* a reference specimen different than that used to conduct the frame calibration, of known thermal expansion preferably in the range of 4.0 – 6.0 µ-strain/°F (7.2 – 10.8 µ-strain/°C), 7.0 ± 0.1 in. (178 ± 2.5 mm) in length and at least a 2 in. (50 mm) diameter.

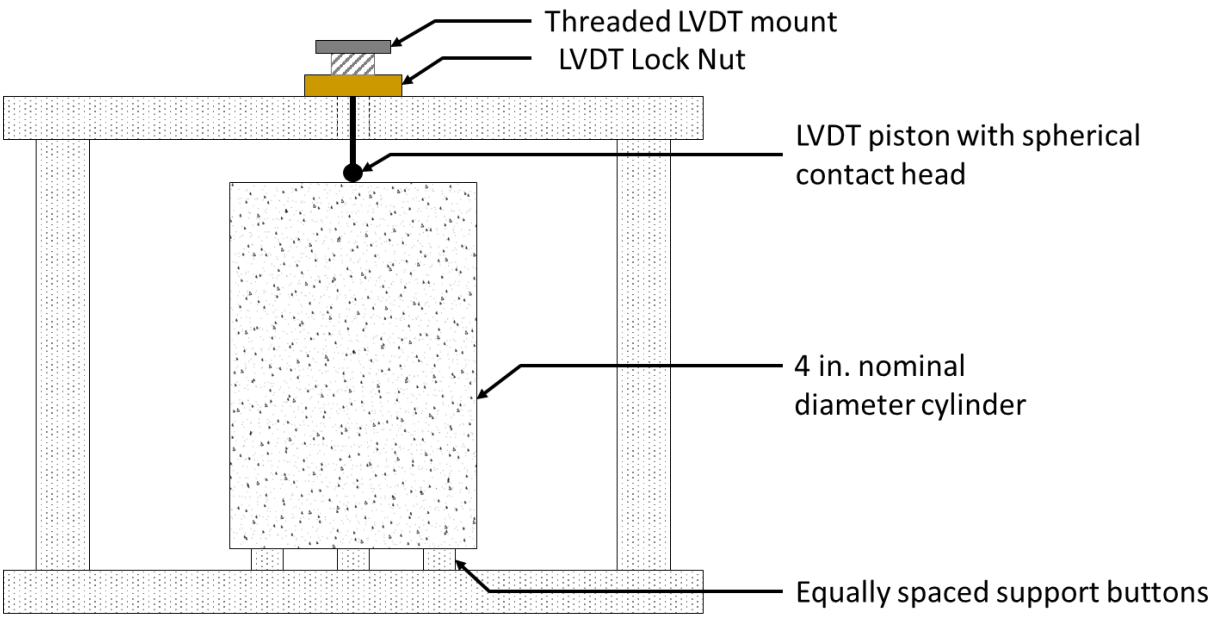


Figure 1
Support Frame

3. TEST SPECIMENS

3.1 Test specimens should consist of a 4 in. (100 mm) nominal diameter core or cylinder.

3.1.1 Obtain cores in accordance with Tex-424-A.

3.1.2 Cast cylinders in accordance with Tex-447-A.

3.2 Saw the specimen perpendicular to the axis, to a standard length of 7 ± 0.1 in. (178 ± 2 mm). Neither end of the test specimen should depart from perpendicularity to the axis by more than 0.5° [approximately equivalent to 0.12 in. per 12 in. (1 mm per 100 mm)].

Note 1—Ensure the cylinder ends are free of gouges, grooves, or indentations that may affect how the cylinder rests on the support buttons or how the LVDT contacts the cylinder.

4. TEST PROCEDURE

4.1 Condition the test specimens by submerging in saturated lime water at 73 ± 4°F (23 ± 2°C) for at least 48 hr. Remove surface moisture with a towel.

4.2 Begin with support frames and water bath at room temperature.

4.3 Remove support frame for each specimen, from the water bath with LVDT attached.

4.4 Remove the specimen from the saturation tank and measure its length at room temperature to the nearest 0.001in. (0.1 mm). Record this length as L. Clean excess lime deposits off before placing in testing frame.

Note 2—All measurements must be complete within five minutes after removing the specimen from the saturation tank.

4.5 Place the specimen in the support frame. Ensure that the LVDT tip seats against the upper end of the specimen, and the lower end of the specimen seats firmly against the support buttons. Be careful not to bang or jar the LVDT tip.

4.6 Use the threaded mount and lock nut to secure the LVDT in the proper position.

4.7 Place the support frame in the tank and begin testing. Take readings of the temperature, LVDT (displacement), and the time at 1 min. intervals for the entire duration of the test.

4.8 Set the temperature of the water bath to 10 ± 1°C (50 ± 2°F). Maintain this temperature for 1 hr.

4.9 Set the temperature of the water bath to 50 ± 1°C (122 ± 2°F) and allow a minimum of 2 hr. for the water to reach the set temperature. Maintain this temperature for 1 hr.

Note 3—Perform tests to verify the length of time needed to achieve the 50°C (72°F) temperature swing for the specific model of water bath used.

4.10 Set the temperature of the water bath to 10 ± 1°C (50 ± 2°F) and allow a minimum of 2 hr. for the water to reach the set temperature. Maintain this temperature for 1 hr.

4.11 This completes one test cycle. Refer to Figure 2 for graphical representation of a test cycle.

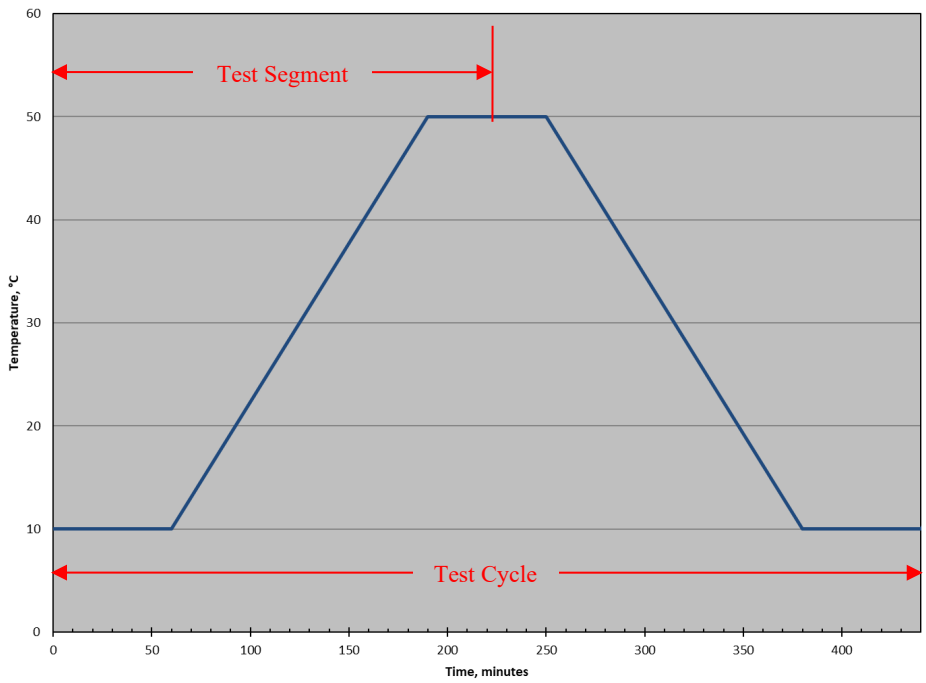


Figure 2
Test Segment and Test Cycle

4.12 Continue cycling between set temperatures of 10 ± 1°C (50 ± 2°F) and 50 ± 1°C (122 ± 2°F), as described in Sections 4.9 and 4.10, until the difference between the CTE value calculated as per Section 5.1 for the decreasing temperature period and the increasing temperature period is less than or equal to 0.17 µ-strain/°F (0.30 µ-strain/°C).

4.13 Report the specimen CTE as the average of the last two segment CTE values.

5. CALCULATIONS

5.1 For each test specimen, plot temperature vs. displacement. Only use increasing or decreasing temperature points. Do not use temperature-maintaining points. Perform regression analysis on the temperature and displacement data. For each specimen, there will be two lines: one for the increasing temperature period and one for the decreasing temperature period. These lines should be approximately parallel to one another. (See Figure 3 for an example) The R^2 value should be greater than 0.999. Find the slope of each line, M. Use the following equations to compute the CTE value.

English Units:

$$CTE = \frac{M}{L*1.8} + C_f$$

Metric Units:

$$CTE = \frac{M}{L} + C_f$$

Where:

CTE = coefficient of thermal expansion value,

M = average slope of falling and rising lines, in./°F (mm/°C),

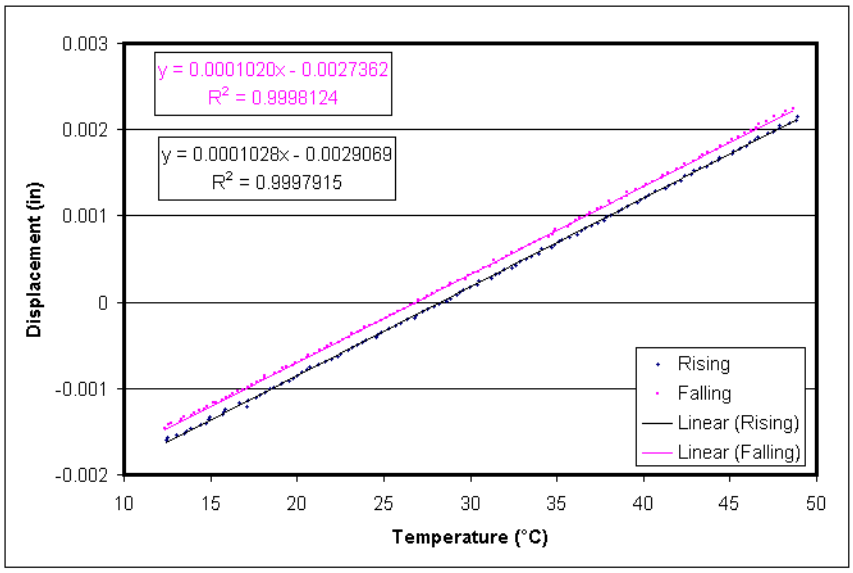
L = length of specimen, in. (mm),

1.8 = conversion factor required to obtain results in terms of °F, and

C_f = correction factor for support frame.

Coefficient of Thermal Expansion

Sample: Frame B With Cal Cyl B



Based on Temperature Cycle Number: 3
LVDT Frame Correction Factor: -0.47

Regression Statistics:

Rising Temperature:		**Falling Temperature:**	
Slope	0.0001028 in/°C	Slope	0.0001020 in/°C
Intercept	-0.0029069	Intercept	-0.0027362
R^2	0.9997915 **OK**	R^2	0.9998124 **OK**
COTE:	7.687 µ-strain/°F	COTE:	7.623 µ-strain/°F

Difference: 0.063 **OK**
Average: 7.655

Overall COTE: 7.66 µ-strain/°F

Figure 3
Example Calculations

6. REPORT

6.1 Include the following in the report:

- identification number,
- age of specimens,
- length of specimen to the nearest 0.001 in. (0.025 mm),
- type of aggregate in specimen, if known,
- individual CTE value, to three significant digits (e.g., CTE = 7.66 x 10^{-6}/°F), and
- average CTE value for all specimens, and record to two significant digits (e.g., CTE = 7.7 x 10^{-6}/°F).

7. PROCEDURE FOR DETERMINING FRAME CORRECTION FACTOR

7.1 Perform the procedure for determining the frame correction factor annually.

7.2 Conduct a test as described in Article 4., "Test Procedure," using the reference specimen as the test specimen.

7.3 Calculate the frame correction factor in accordance with Article 8., "Calculations for Determining Frame Correction Factor."

7.4 Record and enter the frame correction factor to use in calculations for subsequent tests.

8. CALCULATIONS FOR DETERMINING FRAME CORRECTION FACTOR

8.1 Plot temperature vs. displacement. Only use increasing or decreasing temperature points. Do not use temperature maintaining points. Perform regression analysis. There will be two lines: one for the increasing temperature period and one for the decreasing temperature period. The lines should be parallel to one another (see Figure 3). Find the R^2 value for each line. The R^2 value should be greater than 0.999. Find the slope of each line, M. Use the following equations to compute the CTE value for the reference bar.

For English units use:

$$CTE_R = \frac{M}{L * 1.8}$$

For metric units use:

$$CTE_R = \frac{M}{L}$$

8.2 Calculate the correction factor (C_f) and record to three significant digits:

$$C_f = K - CTE_R$$

Where:

K = known coefficient of thermal expansion of reference bar.

9. PROCEDURE FOR SYSTEM VERIFICATION

9.1 Perform the procedure for system verification monthly.

9.2 Conduct a test as described in Article 4., "Test Procedure," using the verification specimen as the test specimen.

9.3 Compare the CTE test result of the verification specimen to the known CTE.

9.4 The final CTE of the verification specimen must be within ±0.1 µ-strain/°F of the known CTE for successful verification.

9.5 When the difference between the measured CTE and the known CTE of the verification specimen is greater than 0.1 µ-strain/°F, recalibrate the LVDT, determine the correction factor as described in Article 7., "Procedure for Determining Frame Correction Factor," of this method, and repeat system verification.

10. ARCHIVED VERSIONS

10.1 Archived versions are available.

ANNEX A—COARSE AGGREGATE PIT TESTING

A.1 STANDARD MIX DESIGN

A.1.1 Materials—Use the following materials for concrete mixtures used for coarse aggregate pit testing.

A.1.1.1 Fine Aggregate—Use a natural, siliceous or siliceous and limestone concrete sand listed in the Concreted Rated Source Quality Catalog.

Note 4—Siliceous sands are material code 062. Siliceous and limestone sands are material code 063.

A.1.1.2 Coarse Aggregate—Sieve the coarse aggregate and recombine in accordance with the requirements in Table 1.

Table 1
Standard Gradation

Sieve Size		Percent by weight of total coarse aggregate content
Passing	Retained	
1"	3/4"	22
3/4"	1/2"	46
1/2"	#4	32

A.1.1.3 Cement—Use a Type I/II or IL Portland cement listed on the Hydraulic Cement MPL.

A.1.1.4 Water—Use a municipal water source.

A.1.2 Mix Design—Use the following design characteristics:

A.1.2.1 Coarse Aggregate Factor (CAF)—0.75

A.1.2.2 Cement content—564 lb./cy

A.1.2.3 W/cm ratio—0.44 (adjusted for aggregate moisture and absorption)

A.2 SPECIMEN PREPARATION AND TESTING

A.2.1 Use the standard mix design to mix concrete in accordance with the standard practice for making and curing concrete test specimens in the laboratory in accordance with ASTM C192.

A.2.2 Prepare a minimum of two 4 in. diameter cylinders and cure in accordance with ASTM C192.

A.2.3 Test one of the prepared cylinders at 28 days in accordance with this procedure.